

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 2 Solutions

1. (a) For $\mathbf{A}$, $\left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{Elimination}]{\text{Gauss-Jordan}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right)$

Since the RREF of $\mathbf{A}$ is the identity matrix (the left side of the resulting matrix), hence $\mathbf{A}$ is non-singular and $\mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$ (the right side of the resulting matrix).

For $\mathbf{B}$, $\left(\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 2 & 4 & 1 & 0 & 1 & 0 \\ -4 & 2 & -9 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 1 & 0 \\ 0 & 0 & 0 & -2 & 1 & 1 \end{array} \right)$

Since there is a zero row in the REF of $\mathbf{B}$ (the left side of the resulting matrix), hence $\mathbf{B}$ is singular.

Note: It is not necessary to do a complete Gauss Jordan elimination if there is already a zero row on the left side.

For $\mathbf{C}$, $\left(\begin{array}{ccc|ccc} 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\text{Elimination}]{\text{Gauss-Jordan}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{6} & -\frac{1}{3} \\ 0 & 1 & 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 1 & 0 & \frac{1}{3} & \frac{2}{3} \end{array} \right)$

Since the RREF of $\mathbf{C}$ is the identity matrix (the left side of the resulting matrix), hence $\mathbf{C}$ is non-singular and $\mathbf{C}^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} & -\frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{pmatrix}$.

- (b) Note that $\mathbf{A}$ is a symmetric matrix, so $\mathbf{A}^T = \mathbf{A}$ and the inverses of $\mathbf{A}^T$ is the same as $\mathbf{A}^{-1}$ in part (a).

Since $\mathbf{B}$ is singular, $\det(\mathbf{B}) = \det(\mathbf{B}^T) = 0$. So $\mathbf{B}^T$ is also singular.

We can find the inverse of $\mathbf{C}^T$ by taking the transpose of $\mathbf{C}^{-1}$ found in part (a).

$$(\mathbf{C}^T)^{-1} = (\mathbf{C}^{-1})^T = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} & -\frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{pmatrix}^T = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ -\frac{1}{6} & \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

- (c) Since $\mathbf{A}$ and $\mathbf{C}$ are non-singular, the homogeneous systems $\mathbf{A}\mathbf{x} = \mathbf{0}$ and $\mathbf{C}\mathbf{x} = \mathbf{0}$ have only the trivial solution.

$\mathbf{B}$ is singular, so the homogeneous system $\mathbf{B}\mathbf{x} = \mathbf{0}$ has non-trivial solutions.

The row echelon form of $\mathbf{B}$ can be obtained from part (a): $\left(\begin{array}{ccc} -1 & 3 & -4 \\ 0 & 10 & -7 \\ 0 & 0 & 0 \end{array} \right)$.

By back substitution and introducing parameter for variable z , we get the general solution for $\mathbf{B}\mathbf{x} = \mathbf{0}$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{19}{10}t \\ \frac{7}{10}t \\ t \end{pmatrix} \text{ or more simply } \begin{pmatrix} -19t \\ 7t \\ 10t \end{pmatrix}$$

for some parameter t .

- (d) $\mathbf{AB}\mathbf{x} = \mathbf{0}$ has non-trivial solutions.

Explanation:

Since we know $\mathbf{B}\mathbf{x} = \mathbf{0}$ has a non-trivial solution $\mathbf{u}$, we have $\mathbf{B}\mathbf{u} = \mathbf{0}$. So $(\mathbf{AB})\mathbf{u} = \mathbf{A}(\mathbf{B}\mathbf{u}) = \mathbf{0}$.

So $\mathbf{u}$ is also a non-trivial solution of $\mathbf{AB}\mathbf{x} = \mathbf{0}$.

Alternatively, since $\mathbf{B}$ is singular, so $\det(\mathbf{B}) = 0$. Then

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}) = 0.$$

This means $\mathbf{AB}$ is singular. So $\mathbf{AB}\mathbf{x} = \mathbf{0}$ has non-trivial solution.

$\mathbf{AC}\mathbf{x} = \mathbf{0}$ has no non-trivial solution.

Explanation:

Since both $\mathbf{A}$ and $\mathbf{C}$ are non-singular, the product $\mathbf{AC}$ is also non-singular.

So we know $\mathbf{AC}\mathbf{x} = \mathbf{0}$ has only the trivial solution.

2. (a) Performing Gaussian elimination, we obtain the row-echelon form of $\mathbf{A}$:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ -1 & 3 & 6 \\ 2 & 1 & 0 \\ 3 & 1 & -1 \end{pmatrix} \xrightarrow{G.E.} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

So rank $\mathbf{A} = 3$.

Performing Gaussian elimination, we obtain the row-echelon form of $\mathbf{B}$:

$$\begin{pmatrix} 2 & 1 & 4 & 1 & 2 \\ 4 & 2 & 2 & 3 & 2 \\ 2 & 1 & -2 & 2 & 0 \\ 6 & 3 & 6 & 4 & 4 \end{pmatrix} \xrightarrow{G.E.} \begin{pmatrix} 2 & 1 & 4 & 1 & 2 \\ 0 & 0 & -6 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

So rank $\mathbf{B} = 2$.

Performing Gaussian elimination, we obtain the row-echelon form of $\mathbf{C}$:

$$\begin{pmatrix} 1 & 4 & 5 & 8 \\ -1 & 4 & 3 & 0 \\ 2 & 0 & 2 & 1 \end{pmatrix} \xrightarrow{G.E.} \begin{pmatrix} 1 & 4 & 5 & 8 \\ 0 & 8 & 8 & 8 \\ 0 & 0 & 0 & -7 \end{pmatrix}$$

So rank $\mathbf{C} = 3$.

- (b) Input the three matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ into MATLAB, apply the command
 $\gg \text{rank}(\mathbf{A}), \text{rank}(\mathbf{B}), \text{rank}(\mathbf{C})$.
- (c) Since $\mathbf{A}$ is a 5×3 matrix and $\text{rank } \mathbf{A} = 3$, so $\mathbf{A}$ is full rank.
 Since $\mathbf{B}$ is a 4×5 matrix and $\text{rank } \mathbf{B} < 4$, so $\mathbf{B}$ is not full rank.
 Since $\mathbf{C}$ is a 3×5 matrix and $\text{rank } \mathbf{C} = 3$, so $\mathbf{C}$ is full rank.
- (d) Observe that in the row echelon form of $\mathbf{C}$, column 3 is non-pivot (no leading entry). If we add a row with leading entry at column 3, for example $(0 \ 0 \ 1 \ 0)$, the resulting 4×4 matrix will have a row echelon form with 4 non-zero rows (by swapping $(0 \ 0 \ 1 \ 0)$ with the third row of the row echelon form of $\mathbf{C}$).
- (e) Since matrix $\mathbf{A}$ has only three columns, no matter what row we add to $\mathbf{A}$, the rank will be bounded by 3. Since $\text{rank}(\mathbf{A}) = 3$, it is not possible for the resulting matrix to have rank larger than 3.
3. (i) $\det(\mathbf{A}) = 1513^3 \neq 0$ (product of diagonal entries), so $\mathbf{A}$ is non-singular.
 $\det(\mathbf{B}) = 0$ (identical columns), so $\mathbf{B}$ is singular.

(ii) $\det\left(\frac{1}{1513}\mathbf{A}\right) = \frac{1}{1513^3} \det(\mathbf{A}) = 1.$

- (iii) $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}) = 0$. So $\mathbf{AB}$ is singular.
 This implies $\mathbf{ABx} = \mathbf{0}$ has non-trivial solutions.

4. (i) The inverse matrix is given by $\begin{pmatrix} 1 & -1 & 0 \\ -3 & 2 & 1 \\ \frac{3}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix}$ (either by hand GJE, or use MATLAB).
- (ii) Let x, y, z be the number of units of types A, B, C to be produced respectively. Then

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ -3 & 2 & 1 \\ \frac{3}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 100 \\ 60 \\ 200 \end{pmatrix} = \begin{pmatrix} 40 \\ 20 \\ 20 \end{pmatrix}.$$

- (iii) Let L be the number of units of labour available for the production. Then

$$\begin{pmatrix} 1 & -1 & 0 \\ -3 & 2 & 1 \\ \frac{3}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} L \\ 60 \\ 200 \end{pmatrix} = \begin{pmatrix} L - 60 \\ -3L + 320 \\ \frac{3}{2}L - 130 \end{pmatrix}.$$

We require the number of units to be produced to be non-negative integers.
 i.e.

$$L - 60 \geq 0 \Rightarrow L \geq 60;$$

$$-3L + 320 \geq 0 \Rightarrow L \leq 320/3 \approx 106;$$

$$\frac{3}{2}L - 130 \geq 0 \Rightarrow L \geq 260/3 \approx 87.$$

So the smallest number of units is 87 and largest number of units is 106.

5. (i) The plane is represented by $ax + by + cz = 0$ (*). As the two vectors $(1, 1, 1), (3, 0, 1)$ lie on the plane, we have:
 $x = 1, y = 1, z = 1$ satisfy the equation (*). So we have $a + b + c = 0$;
 $x = 3, y = 0, z = 1$ satisfy the equation (*). So we have $3a + c = 0$.
Solving these two equations, we get $c = t, a = -t/3, b = -2t/3$, where t is a parameter. Taking $t = -3$, we get a particular solution: $a = 1, b = 2, c = -3$.
Hence the linear equation representing the plane is $x + 2y - 3z = 0$.
- (ii) Substituting $x = 24, y = 9, z = 14$ into the equation in (i), we see that $24 + 2(9) - 3(14) = 0$. So the equation is satisfied. Hence the vector $(24, 9, 14)$ lies on the plane.
- (iii) We compute the dot products:
 $(1, 2, -3) \cdot (1, 1, 1) = 1 + 2 - 3 = 0$ and $(1, 2, -3) \cdot (3, 0, 1) = 3 + 0 - 3 = 0$.
Hence $\mathbf{v}$ is orthogonal to both $(1, 1, 1)$ and $(3, 0, 1)$.
Yes. $\mathbf{v}$ is orthogonal to every vector in the plane.
By definition, every vector in the plane has the form $a(1, 1, 1) + b(3, 0, 1)$.
The dot product $\mathbf{v} \cdot (a(1, 1, 1) + b(3, 0, 1)) = a(\mathbf{v} \cdot (1, 1, 1)) + b(\mathbf{v} \cdot (3, 0, 1)) = a \times 0 + b \times 0 = 0$.
Hence $\mathbf{v}$ is orthogonal to every vector in the plane.
6. (a) (i) Distance: $\|\mathbf{u} - \mathbf{v}\| = \sqrt{20}$,
Angle: $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \times \|\mathbf{v}\|} = \frac{-4}{\sqrt{20}}$, so $\theta = \cos^{-1}(\frac{-4}{\sqrt{20}}) \approx 153.4^\circ$.
- (ii) Since $\mathbf{u} \cdot \mathbf{v} \neq 0$, $\mathbf{u}$ and $\mathbf{v}$ are not orthogonal.
- (b) (i) $\mathbf{u}, \mathbf{v}$ distance: $\|\mathbf{u} - \mathbf{v}\| = \sqrt{27}$,
 $\mathbf{u}, \mathbf{v}$ angle: $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \times \|\mathbf{v}\|} = 0$, so $\theta = 90^\circ$.
 $\mathbf{u}, \mathbf{w}$ distance: $\|\mathbf{u} - \mathbf{w}\| = \sqrt{19}$,
 $\mathbf{u}, \mathbf{w}$ angle: $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\| \times \|\mathbf{w}\|} = 0$, so $\theta = 90^\circ$.
 $\mathbf{v}, \mathbf{w}$ distance: $\|\mathbf{v} - \mathbf{w}\| = \sqrt{12}$,
 $\mathbf{v}, \mathbf{w}$ angle: $\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \times \|\mathbf{w}\|} = \frac{3}{\sqrt{13}\sqrt{5}} = 0.3721$, so $\theta = 68.15^\circ$.
- (ii) Since $\mathbf{u} \cdot \mathbf{v} = 0$, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.
Since $\mathbf{u} \cdot \mathbf{w} = 0$, $\mathbf{u}$ and $\mathbf{w}$ are orthogonal.
Since $\mathbf{v} \cdot \mathbf{w} \neq 0$, $\mathbf{v}$ and $\mathbf{w}$ are not orthogonal.
- (iii) Set up the vector equation $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$ to solve for a, b . Convert this to a linear system with augmented matrix

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 2 & -3 & -1 \\ 3 & 2 & 0 \end{array} \right) \xrightarrow{GJE} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right)$$

As the system is inconsistent, $\mathbf{w}$ is not a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

- (iv) From (iii), $\mathbf{w}$ is not a linear combination of $\mathbf{u}$ and $\mathbf{v}$.
Hence $\text{span}\{\mathbf{u}, \mathbf{v}\} \neq \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

- (c) (i) $\mathbf{u}, \mathbf{v}$ distance: $\|\mathbf{u} - \mathbf{v}\| = \sqrt{14}$,
 $\mathbf{u}, \mathbf{v}$ angle: $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \times \|\mathbf{v}\|} = 0$, so $\theta = 90^\circ$.
 $\mathbf{u}, \mathbf{w}$ distance: $\|\mathbf{u} - \mathbf{w}\| = \sqrt{10}$,
 $\mathbf{u}, \mathbf{w}$ angle: $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\| \times \|\mathbf{w}\|} = \frac{4}{\sqrt{4}\sqrt{14}} = 0.5345$, so $\theta = 57.7^\circ$.
 $\mathbf{v}, \mathbf{w}$ distance: $\|\mathbf{v} - \mathbf{w}\| = 2$,
 $\mathbf{v}, \mathbf{w}$ angle: $\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \times \|\mathbf{w}\|} = \frac{10}{\sqrt{10}\sqrt{14}} = 0.8452$, so $\theta = 32.3^\circ$.
- (ii) Since $\mathbf{u} \cdot \mathbf{v} = 0$, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.
Since $\mathbf{u} \cdot \mathbf{w} \neq 0$, $\mathbf{u}$ and $\mathbf{w}$ are not orthogonal.
Since $\mathbf{v} \cdot \mathbf{w} \neq 0$, $\mathbf{v}$ and $\mathbf{w}$ are not orthogonal.
- (iii) Set up the vector equation $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$ to solve for a, b . Convert this to a linear system with augmented matrix

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ -1 & 1 & 0 \\ 1 & 1 & 2 \\ -1 & 2 & 1 \end{array} \right) \xrightarrow{GJE} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

As the system is consistent, $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

- (iv) From (iii), $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.
Hence $\text{span}\{\mathbf{u}, \mathbf{v}\} = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.